

JACK GILBERT J

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📍 Krishnagiri, India

🌐 Jack Gilbert

🎓 EDUCATION

Bachelor of Technology in Computer Science Engineering

2021 – 2025

Karunya Institute Of Technology and Science — Coimbatore

🏢 EXPERIENCE

Genau Extrusions — NPD Tool Design Engineer — Hosur

Dec 2025 – Present

- Design forging tools and components using **AutoCAD, SolidWorks, and Simufact Forming**, focusing on tool life and manufacturability.
- Work on **tool-life optimization, design improvements, and standardization** for NPD tooling.
- Develop **AutoLISP automation** using AutoCAD APPLOAD features to automate repetitive tool-design and drafting tasks.
- Apply parameter-based design and process automation to improve design efficiency and consistency.
- Collaborate with engineering and production teams to address design and manufacturing requirements.

🔧 MAJOR UNIVERSITY PROJECTS

Real-Time Coconut Tree Disease Detection Using YOLOv8 and Explainable AI

May 2025

- Developed a real-time disease detection system using **YOLOv8**, capable of identifying multiple coconut tree diseases such as bud rot and gray leaf spot from image and video inputs.
- Integrated Explainable AI (**EigenCAM**) to generate visual heatmaps, providing farmers with transparent insights into AI predictions for better trust and decision-making.
- Built a **Streamlit-based user interface** supporting image uploads, webcam feeds, and RTSP streams for remote monitoring.
- Achieved effective performance on low-resource devices and included automated disease management recommendations to help reduce crop loss.

Lane Lines Detection using Python and OpenCV

Mar 2023

- Developed an image and video-based lane detection system using **OpenCV and Python** to identify road lane boundaries.
- Implemented preprocessing steps including color filtering, grayscale conversion, Gaussian blur, and Canny edge detection.
- Applied region of interest masking and Hough Line Transform to detect and visualize lane lines accurately.
- Averaged and extrapolated lane lines for robustness across diverse road scenarios.
- Demonstrated effectiveness on static images and video clips, contributing to autonomous driving and driver assistance technologies.

🌟 CERTIFICATIONS

Introduction to Data Science, Cisco Networking Academy

Sep 2023

- Completed 30 hours of lectures and tutorials covering foundational concepts in data science, including data wrangling, visualization, and statistical analysis.
- Gained hands-on experience with Python, Jupyter Notebooks, and real-world datasets.

Master the Fundamentals and Advanced Concepts, Scaler

Oct 2023

- Completed a course covering Data Structures & Algorithms, System Design, SQL, and Low-Level Design.
- Gained hands-on experience with Java, Spring Boot, React, and Redux through a capstone project.

Google Cybersecurity, Coursera

Apr 2024

- Completed 8 courses covering threat detection, network security, SIEM tools, and risk management.
- Gained hands-on experience with Wireshark, Linux, SQL, and Splunk.

Google Data Analytics, Coursera

Apr 2024

- Completed 8 courses focusing on data cleaning, analysis, and visualization using SQL, Excel, R, and Tableau.
- Applied data analytics processes to real-world case studies, including data preparation, hypothesis testing, and dashboard creation.

Course on Ethical Hacking, Cisco Networking Academy

Oct 2024

- Acquired hands-on skills in ethical hacking methodologies, reconnaissance, network scanning, and system exploitation.
- Completed tutorials covering penetration testing, vulnerability assessment, and common attack vectors.

Meta Android Developer, Coursera

Oct 2024

- Completed 8 self-paced courses covering Kotlin programming, Android Studio, UI/UX design, and mobile architecture.
- Developed a food ordering app demonstrating proficiency in MVVM architecture, LiveData, and API integration.

Course on Intro to Machine Learning, Kaggle

Oct 2023

Course on Data Visualization, Kaggle

Sep 2023

🔗 SKILLS

Computer Languages — Python, Java, Matlab, SQL, C, Linux

Data Science — Machine Learning, Deep Learning, Data Engineering, Data Analytics, Statistics

Engineering Tools — AutoCAD, SolidWorks, Simufact Forming, AutoLISP

Data Visualization — Excel, Tableau, Power BI, Google Analytics Studio, Matplotlib, Seaborn

Soft Skills — Solution Centric Thinker, Creative, Resilient, Promptitude, Self Monitoring, Adaptive

♥ INTERESTS

Data Analytics — Machine Learning — Deep Learning — Data Engineering — Data Visualization

Data Management — Business Intelligence — Market Research — Data Scavenging